CSE558 · DATA SCIENCE — LECTURE 8

Goodness of Fit

The liver-fat prediction gets its answer, two new categorical tests get a first look, and one long-standing question what can be quantified from samples.

Supratim · IIIT-Delhi · Monsoon 2026

WHERE WE LEFT OFF

Lecture 7, in four moves

Two means

z if σ is known, t if it isn't. Pooled or Welch depending on the variances.

Two proportions

Pool the rate — but only when both samples share one population.

Correlation

t-test on r, with n − 2 degrees of freedom.

Two variances

The F-test. A ratio, not a difference over an error.

… and one comparison we deliberately left hanging.

BACK TO THE UNFINISHED BUSINESS

Does liver fat % differ between men and women?

Men (n = 13) — group 1

Mean ≈ 20.95
Std dev ≈ 26.71

(13.3, −6, 20, 8, 14, 19, 18, 25, 16, 24, 15, 1, 105)

Women (n = 10) — group 2

Mean ≈ 22.29
Std dev ≈ 5.32

(22, 16, 21.7, 21, 30, 26, 12, 23.2, 28, 23)

H_0: $\mu_1 = \mu_2$ H_1: $\mu_1 \neq \mu_2$ $\alpha = 0.05$, two-sided

Test mean & variance before pooled statistics.

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s\sqrt{1/n_1 + 1/n_2}}$$

Pooled $s^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2]/(n_1 + n_2 - 2)$

BEFORE YOU POOL

Check first: are the variances even equal?

Before trusting a pooled t-test, run the F-test on the two variances.

$$F = s_1^2 / s_2^2 = 47.46 / 28.30 = 1.68$$

d.f. = (10, 9) critical F (two-sided, α = 0.05) = 3.96

1.68 < 3.96 — do not reject.

Pooling would have been the wrong move, if the variance or the means are significantly different. Validate this before a single t-value is computed.

TWO NEW QUESTIONS

From comparing groups to comparing counts

Every test since Lecture 5 has asked about a mean, a proportion, or a spread — always a single number pulled from continuous measurements.

Today's two previews ask a different kind of question: does this data, sitting in categories and counted up, look the way you expected?

PREVIEW ONE

χ^2-test for goodness of fit

Use it when: you're comparing an observed split across categories to a claimed one. Data are counts, not measurements. d.f. = k − 1, where k is the number of categories.

from sample → O_i ; Expected → E_i

$$\chi^2 = \sum_{i=1}^{k} (O_i - E_i)^2 / E_i$$

O_i = observed count in category i *E_i = expected count in category i*

H_0: The population follows a certain distribution
H_1: It doesn't.

WORKED EXAMPLE

Does the canteen's claim hold up?

	Level of significance α							
Two-sided One-sided	0.20 0.10		0.10 0.05		0.05 0.025		0.01 0.005	
ν	a	b	a	b	a	b	a	b
1	0.016	2.71	39.10^{-4}	3.84	98.10^{-5}	5.02	16.10^{-5}	6.63
2	0.21	4.61	0.10	5.99	0.05	7.38	0.02	9.21
3	0.58	6.25	0.35	7.81	0.22	9.35	0.11	11.34
4	1.06	7.78	0.71	9.49	0.48	11.14	0.30	13.28
5	1.61	9.24	1.15	11.07	0.83	12.83	0.55	15.09

The canteen claims all 5 lunch items are ordered equally often. A day's log of 200 orders:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

Item 1	Item 2	Item 3	Item 4	Item 5	Total
35	45	38	52	30	200

Item 1	Item 2	Item 3	Item 4	Item 5	Total
40	40	40	40	40	200

$$\chi^2 = \frac{25 + 25 + 4 + 144 + 100}{40}$$

ONE STEP FURTHER

Does this look like a Poisson process?

	Level of significance α							
Two-sided One-sided	0.20 0.10		0.10 0.05		0.05 0.025		0.01 0.005	
ν	a	b	a	b	a	b	a	b
1	0.016	2.71	39.10^{-4}	3.84	98.10^{-5}	5.02	16.10^{-5}	6.63
2	0.21	4.61	0.10	5.99	0.05	7.38	0.02	9.21
3	0.58	6.25	0.35	7.81	0.22	9.35	0.11	11.34
4	1.06	7.78	0.71	9.49	0.48	11.14	0.30	13.28
5	1.61	9.24	1.15	11.07	0.83	12.83	0.55	15.09

A company's support desk was staffed around the assumption of x tickets an hour. They have a sample of 50 hours as follows. At α = 0.05, does this justify a change in staffing?

0	1	2	3	4
7	10	17	8	8

0	1	2	3	4+
27.06				

→ 4 or more defects.

$\lambda = \frac{100}{50} = 2$

$P_\lambda(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

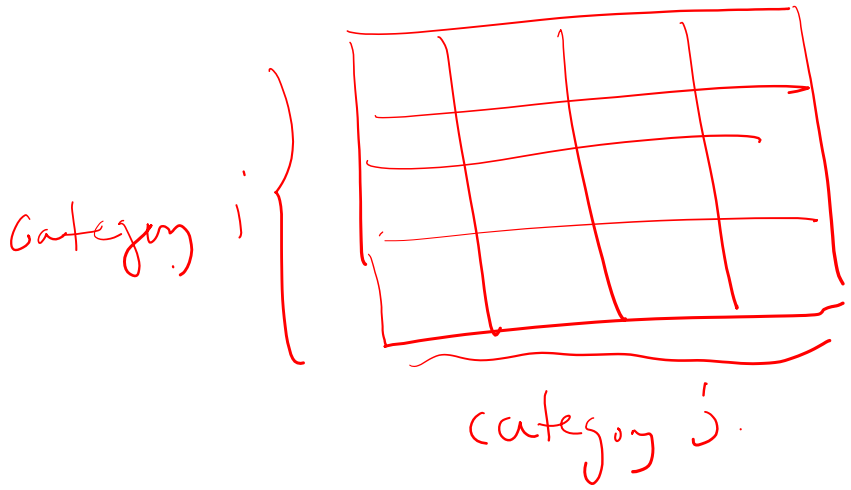
PREVIEW TWO

χ^2-test for independence

Use it when: two categorical variables, one sample, cross-classified — asking whether they're associated. d.f. = (r − 1)(c − 1), for r rows and c columns.

$$\chi^2 = \sum_{i,j} \left(O_{i,j} - E_{i,j}\right)^2 / E_{i,j}$$

Same formula as goodness of fit — only E_{ij} now comes from row totals × column totals ÷ grand total.



WORKED EXAMPLE

Does language preference depend on year?

	Python	C++	Java	Total
3rd year	55	30	35	120
4th year	25	35	20	80
Total	**80**	**65**	**55**	**200**

	Python	C++	Java	Total
3rd year				120
4th year				80
Total	**80**	**65**	**55**	**200**

$Pr(A \& B) = Pr(A) \cdot P(B)$ if A & B are independent



STUDY MATERIAL

Reference

The Art of Learning from Data: Chapter 10.2, 11 by Agresti, Franklin & Klingenberg

Slides